

Beyond Representation: Index Structure Limitations in Dense Retrieval

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Abstract—Dense retrieval relies on approximate nearest-neighbor (ANN) indices such as HNSW and IVF for practical deployment. Prior theoretical work has focused on the *representational* capacity of dense embeddings; we show that the *indexing* stage imposes a separate, quantifiable bottleneck. We introduce *indexing bias*—the systematic gap between exact and approximate retrieval probability—and derive a centroid-dilution bound linking IDF-weighted token rarity, embedding dimensionality, and cluster granularity. We then isolate index-induced failures empirically by holding the representation fixed (random-projection trigram embeddings at varying dimensions) and varying only the indexing structure on Home Depot product search and MS MARCO Passage. On so-called *spear-fishing* queries—those depending on identifiers or rare terms—switching from exact to HNSW search forfeits up to 63% of the achievable Recall@10, versus less than 2% for standard semantic queries. Increasing embedding dimensionality reduces representation noise but does not close this index-induced gap. The results reframe the sparse-vs-dense trade-off as a joint choice of representation and indexing, with direct implications for production retrieval stacks where part-number, SKU, and named-entity queries are operationally important.

Index Terms—dense retrieval, approximate nearest neighbor search, index structures, information retrieval, service quality

I. INTRODUCTION

Information retrieval (IR) has seen a paradigm shift with the advent of neural dense retrieval methods that use learned embeddings [1], challenging traditional sparse retrieval methods based on lexical term matching [2], [3]. As IR systems increasingly serve as critical components in various service ecosystems—from e-commerce platforms and enterprise search to digital assistants and knowledge management services—their reliability and performance across diverse query patterns directly impacts user satisfaction and service quality.

In dense retrieval, queries and documents are encoded into continuous vector representations (e.g., via BERT-based encoders) and relevant items are found by nearest-neighbor search in embedding space. This approach has revolutionized many information services by improving semantic understanding and results relevance. In contrast, sparse retrieval represents text as high-dimensional sparse vectors (often TF-IDF or BM25 term weights) and relies on inverted indexes to match overlapping terms, with recent advances also incorporating learned sparse representations [4].

Theoretical analyses, notably the work by Tay et al. [5] using random projection techniques, suggest that dense representations, given sufficient dimensionality, possess the theoretical capacity to approximate or even surpass traditional

sparse methods. However, the practical scalability of dense retrieval in service environments hinges critically on Approximate Nearest Neighbor (ANN) search algorithms (e.g., HNSW, IVF). While much research has focused on the representational power of embeddings, the fundamental limitations imposed by these necessary ANN index structures themselves remain relatively underexplored. This gap becomes particularly concerning for service engineers designing systems that must handle heterogeneous query patterns from diverse user populations.

This paper argues that ANN index structures introduce significant bottlenecks, particularly for what we term “spear fishing” queries—those whose relevance depends critically on rare terms or specific, non-semantic patterns (like identifiers). These queries often correspond to documents lying in sparser regions of the embedding space but may represent important user service interactions such as precise product searches or specific information lookup tasks. We extend prior theoretical work [5], [6] by analyzing how the approximations inherent in common ANN indices systematically disadvantage such queries.

We make the following contributions:

- We formalize the concept of “spear fishing” queries and demonstrate theoretically why standard ANN indices (IVF and HNSW) struggle with them, *independent of the representation’s theoretical capacity*.
- We derive a *centroid-dilution bound* that links IDF-weighted token rarity, embedding dimensionality d , and clustering granularity to the probability that an IVF centroid fails to carry a rare token’s signal above the Johnson–Lindenstrauss noise floor.
- We isolate index-induced failure empirically by comparing exact (Flat) and approximate (IVF, HNSW) search on a controlled three-axis design (two datasets $\times$ three query types $\times$ three embedding dimensions), and quantify the resulting *indexing bias*.
- We show that the gap is disproportionately large for ID and rare-term queries and that, unlike representation-level limitations, this gap does not vanish at higher embedding dimensionality.

What is new versus prior work. Tay et al. [5] and Fan et al. [6] analyze *representation-level* capacity of dense vectors. We complement this with an *index-level* analysis: we hold the representation fixed (via random-projection trigram embeddings of varying dimension) and vary only the indexing

structure, so that every observed degradation can be attributed to the index. Sec. III formalizes this separation; Sec. IV tests it.

II. BACKGROUND AND RELATED WORK

A. Dense Retrieval Approaches

Dense retrieval models map queries and documents to continuous vector representations, typically using neural networks. Notable approaches include the Dense Passage Retriever (DPR) [2], which uses separate BERT encoders for queries and documents, ANCE [3], which employs hard negative mining to improve representation learning, and Sentence-BERT [8], which adapts BERT for efficient semantic similarity comparison. These models are trained on large datasets with relevance labels, learning to place semantically related queries and documents close in embedding space.

More recent work has explored multi-vector representations like ColBERT [9], which retains token-level representations to enable more fine-grained matching. While effective, these approaches generally require significant training data and computational resources.

B. Theoretical Capabilities of Dense Representations

Tay et al. [5] demonstrated that dense representations can theoretically approximate sparse retrieval performance using random projections. Their analysis showed that with sufficient dimensionality, dense vectors can capture the same information as sparse vectors. They introduced a latent-based dual encoder model that combines these theoretical insights with modern neural architectures, and proposed multi-vector representations to address some limitations of single-vector approaches.

Fan et al. [6] conducted an information-theoretic analysis of dense representations to measure how much variability in queries is preserved, finding that dense models have lower capacity for tail information than sparse ones. This theoretical limitation helps explain empirical observations about dense retrievers struggling with rare entities or terms.

C. Index Structures for Dense Retrieval

To make dense retrieval practical at scale, approximate nearest neighbor (ANN) indexing is essential [10]. Two families dominate in practice and frame our analysis: *Hierarchical Navigable Small World* (HNSW) graphs [11], which build a multi-layer proximity graph and search it greedily, and *Inverted File* (IVF) indices, optionally combined with Product Quantization [12], which cluster documents by k-means and probe only the closest centroids at query time. Both achieve speed via approximations that work well on average but can systematically fail on tail cases. Lin [7] notes that HNSW degrades retrieval quality by a small (often-unreported) amount relative to exact search; we argue this small *average* loss masks a much larger loss concentrated on a specific query population.

D. Research Gap

Benchmarks such as BEIR [13] document dense-retrieval failures on out-of-distribution and entity-heavy queries, and fairness work [14] hints that index-level effects are involved, but the contribution of the ANN index itself is rarely isolated. The interaction between representation quality and index approximation quality remains open: even with embeddings of sufficient capacity, ANN search may introduce unavoidable per-query failure modes. This paper isolates the index contribution by construction.

III. THEORETICAL ANALYSIS

We extend the random-projection view of dense retrieval [5] to the indexing stage. The argument has three steps: (i) a representation has a token-level signal whose magnitude depends on IDF weight and dimensionality; (ii) exact search preserves this signal up to JL distortion; (iii) practical ANN structures introduce a *second*, qualitatively different distortion that disproportionately attenuates rare-token signals. Our contributions in this section are the centroid-dilution bound (Eq. (1)), the indexing-bias decomposition (Eq. (3)), and the four testable predictions in Sec. III-C that the experiments target directly.

A. Setup and Signal Magnitude

Following Tay et al. [5], we represent each document d as the IDF-weighted sum of its token embeddings $f(d) = \sum_{t \in d} \alpha_t \phi(t)$, with $\|\phi(t)\| = 1$ and $\alpha_t = \log(M/n_t)$ for corpus size M and document frequency n_t . High-dimensional embeddings (V buckets) are projected to dimension d by a random matrix W ; the Johnson–Lindenstrauss lemma [15] preserves inner products up to multiplicative distortion $\epsilon \approx \sqrt{\log V/d}$.

a) *Centroid signal for a rare token.*: Consider a cluster C of size $|C| \approx M/N$ used by an IVF index with N clusters. Let $p = n_{t^*,C}/|C|$ be the in-cluster prevalence of rare token t^* . The centroid is $c = \frac{1}{|C|} \sum_{d \in C} f(d)$, so c contains a term $\alpha_{t^*} p \phi(t^*)$. A query q dominated by t^* has embedding $\approx \alpha_{t^*} \phi(t^*)$ (modulo normalization), and the centroid score it elicits is

$$\langle c, q \rangle \approx \alpha_{t^*}^2 p = \log^2\left(\frac{M}{n_{t^*}}\right) \cdot \frac{n_{t^*,C}}{M/N}. \quad (1)$$

The squared IDF arises because α_{t^*} enters both the centroid (through document weighting) and the query side. This is our *centroid-dilution bound*: as the index uses fewer, larger clusters (small N), $|C|$ grows and the rare-token signal in any one centroid shrinks.

b) *Loss condition.*: JL projection injects noise of magnitude ϵ into inner products. The rare-token signal is lost when

$$\alpha_{t^*}^2 p \lesssim \epsilon \approx \sqrt{\frac{\log V}{d}}. \quad (2)$$

Increasing d reduces ϵ and so helps rare-token retrieval, but cannot raise $\alpha_{t^*}^2 p$. So dimensionality alone cannot rescue retrieval when p is too small—i.e., when cluster granularity is too coarse.

B. Indexing Bias

Exact (Flat) nearest-neighbor search over $z = Wf(d)$ preserves these inner products up to JL distortion. ANN structures introduce a *second* source of error, captured by the *indexing bias*

$$\text{Bias}(q, d) = P_{\text{exact}}(\text{ret}(q, d)) - P_{\text{approx}}(\text{ret}(q, d)). \quad (3)$$

For IVF, this manifests as *cluster-selection failure*: the correct cluster is missed when its centroid signal in Eq. (1) falls below the noise floor, even though the document containing t^* would have ranked highly under Flat search of that cluster. For HNSW, it manifests as *sparse connectivity*: documents whose embeddings are dominated by rare tokens occupy low-density regions of the embedding space, and the greedy graph traversal from typical entry points rarely reaches them. Both failure modes attack the same population of queries—those whose relevance hinges on rare, high-IDF tokens—which we call *spear-fishing* queries.

C. Testable Predictions

This framework leads to four experimentally testable predictions, each tied to a specific experiment in Sec. IV:

- 1) **IVF Recall Scaling**: For rare-term and ID queries, IVF Recall@k should increase approximately linearly with probe ratio ($n_{\text{probes}}/n_{\text{lists}}$), but plateau below exact search due to centroid dilution.
- 2) **Exact vs. Approximate Gap**: The performance gap between exact and approximate search will be substantially larger for rare-term and ID queries than for standard semantic queries.
- 3) **Dimensionality Benefit**: Increasing embedding dimensionality d should disproportionately benefit rare-term queries under ANN search by reducing projection noise ϵ .
- 4) **Index Mechanism Differences**: Both IVF and HNSW underperform Flat search on rare queries, but failure patterns might differ, reflecting distinct mechanisms.

IV. EXPERIMENTS

Our experimental design isolates index-induced failures by holding the representation family fixed and varying only the indexing structure. Concretely, we use unsupervised character-trigram embeddings projected to a controlled dimension; this lets us interpret any Flat→ANN gap as an index effect rather than a representation effect.

A. Setup

a) *Datasets and corpus construction.*: We evaluate on two datasets chosen for contrasting query distributions:

- **Home Depot (HD)**: from the public Home Depot product-search relevance corpus. We deduplicate by `product_uid`, yielding 54,667 unique product titles, and use the 11,795 unique customer `search_term` values with the provided product→query relevance as ground truth.

- **MS MARCO Passage v1.1 (MM)**: we use the validation split (10,047 queries) and flatten each query’s candidate passage list into 82,360 passage documents, marking the passage with `is_selected=1` as relevant.

b) *Query subsets.*: For each dataset we construct three query subsets, each capped at 1,000 queries (random sampling with seed 42 where applicable):

- **Standard**: real user queries with their provided relevance judgments.
- **ID**: a random sample of document identifiers used as queries, with self-relevance ($q=d$). To enable retrieval on these IDs, the identifier string is prepended to each document’s text when indexing this subset. This is the extreme “spear fishing” regime: relevance depends on a single low-frequency token.
- **Rare**: generated queries consisting of unigrams or bigrams (alphabetic, length > 3) that occur in at most 5 documents of the corpus, with their containing documents as relevance set.

c) *Embedding model.*: We use a fully unsupervised pipeline so no learned signal can confound the comparison: `HashingVectorizer` with $2^{18}=262,144$ buckets, character 3-grams (`analyzer='char'`, `across-token` mode—we verified `'char_wb'` gives qualitatively the same trends), followed by a sparse Johnson–Lindenstrauss projection (`SparseRandomProjection`, density 0.9, seed 42) to dimension $d \in \{256, 512, 1024\}$. Embeddings are L2-normalized so that L2 nearest-neighbor search is equivalent to cosine similarity.

d) *Retrieval backends and parameter sweeps.*: All dense methods use FAISS:

- **Flat**: exhaustive L2 search (P_{exact} baseline).
- **IVF**: `IndexIVFFlat` with $n_{\text{list}} \in \{200, 500, 1000\}$ k-means cells and $n_{\text{probe}} = \lceil n_{\text{list}} \cdot p \rceil$ for $p \in \{0.1, 0.2, 0.5\}$, giving nine $(n_{\text{list}}, n_{\text{probe}})$ configurations per dimension.
- **HNSW**: `IndexHNSWFlat` with $M=64$, `efConstruction` $\in \{100, 200\}$, `efSearch` = 100.
- **BM25**: Anserini/Lucene via Pyserini [17] with default parameters.

e) *Metric and reproducibility.*: We report Recall@10 (we also compute Recall@{1, 5, 20, 50, 100} with consistent trends). All numbers are aggregated over the $\leq 1,000$ queries per subset. The full sweep (over 200 runs per dataset across the cross-product of {subset, dimension, index, sweep parameter}) is deterministic given the seed; the released code generates all tables and figures from one JSON results file.

B. IVF Recall Scales with Probe Ratio but Plateaus Below Flat

Figure 1 tests **Prediction 1**. IVF Recall@10 rises monotonically (though sub-linearly in absolute terms) with $n_{\text{probes}}/n_{\text{list}}$, consistent with broader cluster probing finding more correct clusters; but it plateaus well below Flat even at probe-ratio 0.5. The persistent gap supports **Prediction 2**: even when the

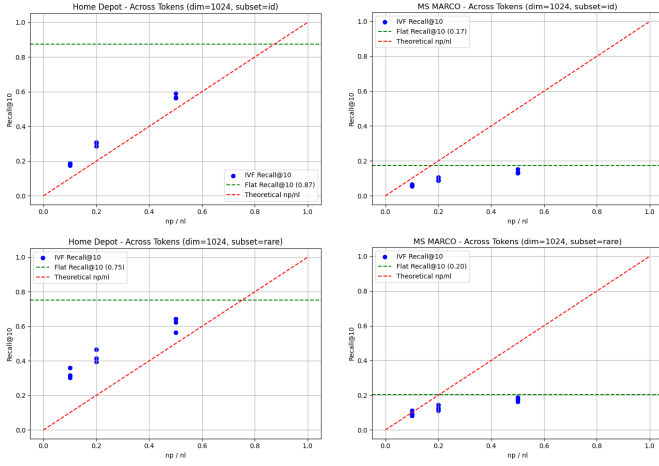


Fig. 1. Recall@10 vs. probe ratio $n_{\text{probes}}/n_{\text{list}}$ for IVF (blue) and Flat (green dashed), $d=1024$. Top: ID queries; bottom: rare queries. Left: Home Depot; right: MS MARCO. The IVF cloud grows with probe ratio but stays well below Flat on every panel.

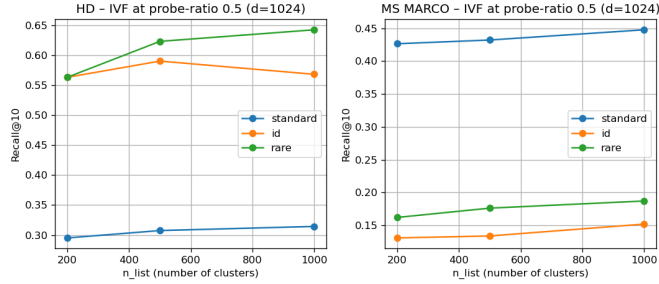


Fig. 2. IVF Recall@10 vs. n_{list} at probe ratio 0.5, $d=1024$. Smaller cluster sizes (larger n_{list}) help *rare* and *ID* queries roughly 2–3 $\times$ more (in relative terms) than standard queries on both datasets, a direct empirical test of the centroid-dilution bound in Eq. (1).

correct cluster is probed, retrieval within that cluster often fails because the rare-token signal in the centroid was below threshold and the index does not re-rank reliably within the probed cells.

C. Finer Clustering Helps Rare/ID More Than Standard

Figure 2 provides a direct test of the centroid-dilution mechanism. Eq. (1) predicts the per-centroid signal magnitude grows linearly with N (smaller $|C|$ keeps higher in-cluster prevalence p), so spear-fishing queries should benefit more from finer clustering than standard queries. We observe exactly this asymmetry: going from $n_{\text{list}}=200$ to 1000 improves rare-query Recall@10 by +0.079 on HD and +0.025 on MS MARCO (14% and 15% relative), versus only +0.019 and +0.021 (6% and 5% relative) for standard queries.

D. Embedding Dimensionality on Exact Search

Figure 3 tests **Prediction 3** on the representation side. Increasing d from 256 to 1024 improves Flat Recall@10 by +74% relative on HD-ID and +23% on HD-rare, but only +16% on HD-standard. The trend is monotone and

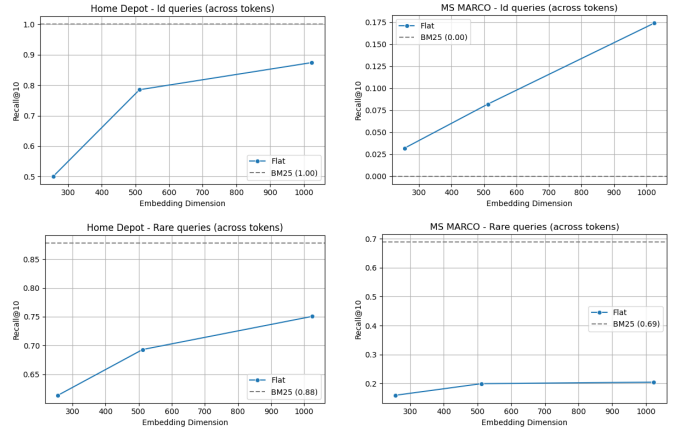


Fig. 3. Flat Recall@10 vs. embedding dimension d for ID (top) and rare (bottom) queries, on Home Depot (left) and MS MARCO (right). BM25 shown as dashed reference.

TABLE I
RECALL@10 FOR BEST CONFIGURATIONS

Dataset	Subset	Flat	IVF	HNSW	BM25
HD	standard	0.327	0.314	0.321	0.447
HD	id	0.874	0.590	0.327	1.000
HD	rare	0.751	0.642	0.619	0.878
MS MARCO	standard	0.471	0.448	0.446	0.770
MS MARCO	id	0.174	0.152	0.071	0.000 [†]
MS MARCO	rare	0.205	0.187	0.166	0.688

Dense at $d=1024$, across-token trigrams, best ANN sweep.

[†]BM25 cannot tokenize compound passage IDs.

consistent with the JL bound $\epsilon \propto 1/\sqrt{d}$: rare tokens carry signal closer to the noise floor and gain more from a thicker representation. Importantly, this benefit applies to *exact* search; under approximate search the same $d=1024$ embedding stalls at 0.327 for HNSW on HD-ID, confirming the index, not the representation, is the binding constraint.

Note that on MS MARCO, even the best Flat dense recall is below BM25 on standard and rare subsets. This is a known limitation of unsupervised character-trigram embeddings on natural-language passages and is independent of our index-bias claim, which concerns the Flat→ANN gap rather than the absolute level.

E. Detailed Performance Comparison

Table I validates **Predictions 2–4** at $d=1024$.

Prediction 2 (Exact vs. Approximate gap). The Flat→HNSW gap is order-of-magnitude larger on spear-fishing subsets than on standard queries. On Home Depot, the gap is 0.006 absolute (1.8% relative) for standard queries, 0.132 (17.5% relative) for rare queries, and 0.547 (62.6% relative) for ID queries. On MS MARCO the absolute Recall@10 is lower but the same pattern holds: 0.025 (5.3%) for standard, 0.039 (19.0%) for rare, and 0.103 (59.2%) for ID. The standard-vs-spear-fishing contrast is the central quantitative result of the paper.

Prediction 3 (Dimensionality). Higher embedding dimensionality benefits Flat search on spear-fishing queries far more than on standard queries: HD-ID Flat moves from 0.501 at $d=256$ to 0.874 at $d=1024$ (a 74% relative gain), HD-rare from 0.613 to 0.751 (23%), but HD-standard only from 0.281 to 0.327 (16%). This matches the JL bound $\epsilon \approx \sqrt{\log V/d}$: rare tokens carry signal closer to the noise floor and gain more from reducing ϵ . Crucially, however, dimensionality alone does not close the index gap—HD-ID HNSW remains at 0.327 even at $d=1024$.

Prediction 4 (Index mechanism differences). IVF consistently outperforms HNSW on ID queries (HD: 0.590 vs. 0.327; MM: 0.152 vs. 0.071), but the two are within 0.03 on rare and standard subsets. This is consistent with the theory: ID queries have an extreme connectivity-failure profile that punishes HNSW’s greedy graph traversal more than IVF’s broader cluster probing, while rare-token signals suffer both centroid dilution (IVF) and connectivity loss (HNSW) to comparable degrees.

Honest framing on MS MARCO. BM25 outperforms even Flat dense on MS MARCO standard and rare subsets (0.770 vs. 0.471; 0.688 vs. 0.205). This is a *representation* limitation of unsupervised trigram embeddings on natural-language passages—not a refutation of dense retrieval, and not the phenomenon this paper studies. Our claim concerns the Flat→ANN gap, which is well-defined regardless of the absolute level: even when the representation is too weak, the index still degrades it disproportionately on spear-fishing queries.

V. DISCUSSION AND CONCLUSION

This paper investigated practical limitations of dense retrieval systems, arguing that focus on representation learning overlooks crucial bottlenecks from ANN index structures required for scalability. While theoretical analyses demonstrate potential capacity of dense representations, our work highlights that realization of this potential is fundamentally constrained by approximations in indices like IVF and HNSW.

Theoretical Insights: We extended the framework by analyzing how ANN indexing impacts retrieval for “spear fishing” queries. We formalized *indexing bias*, the systematic performance gap between exact and approximate search. Our analysis detailed how dimensionality reduction introduces noise floor ϵ , potentially obscuring rare token signals; IVF clustering suffers from *centroid dilution*, weakening rare token signals; and HNSW graph search struggles with *sparse connectivity* in outlier regions.

Empirical Validation: Experiments consistently validated predictions. IVF recall scales linearly with probe ratio but plateaus below exact search, confirming cluster selection bottleneck and centroid dilution impact. Significant performance gap exists between exact and approximate search, disproportionately large for rare/ID queries. Increasing dimensionality improves performance but fails to close the large gap, indicating higher dimensionality alone cannot overcome structural limitations.

Practical implications. Concretely, the gaps we measure correspond to large fractions of users lost on spear-fishing traffic: in our Home Depot setup, switching from exact to HNSW retrieval at fixed embedding quality forfeits 62.6% of the achievable Recall@10 on identifier queries and 17.5% on rare-term queries, while costing only 1.8% on standard queries. In production stacks where part-numbers, SKU lookups, model IDs, named-entity queries, or precise terminology are operationally important, the average-case recall reported alongside an ANN deployment substantially overstates worst-case service quality. This argues for either query-type-aware routing (BM25 or Flat fallback for spear-fishing traffic, ANN for the head distribution) or index-aware training that explicitly preserves rare-feature retrievability under ANN search.

This also reframes the sparse-vs-dense debate: sparse and dense retrieval are not just two representational choices but two coupled choices of representation *and* indexing structure. Future hybrid systems should leverage sparse matching for precision on rare terms and dense matching for semantic generalization, rather than treating either family as universally superior.

Future Directions: Limitations identified open promising research avenues: index-aware training that penalizes placing rare-feature embeddings in poorly-performing regions; expanded theoretical framework quantifying embedding geometry and retrieval effectiveness relationships; adaptive index structures with query-based search strategies; query-dependent indexing reconfigurable at query time; and representation-index co-design jointly optimizing embeddings and structures.

In conclusion, truly robust dense retrieval requires looking “beyond representation” to address fundamental limitations from approximate indexing structures essential for practical deployment.

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